DS	**Session**	Sorting	**Question Information**	Level 1 Challenge 20

Problem Description:

Real estate is property consisting of land and the buildings on it, along with its natural resources such as crops, minerals or water; immovable property of this nature; an interest vested in this (also) an item of real property, (more generally) buildings or housing in general.
There are 'n' flats in the village Nelvayal. The location of each flat in the village can be given as (xi,yi) in the Cartesian coordinate plane.

There are "hi" persons living in the i-th flat. Central electricity authority of the village is set to built a wire line across the village.

The wire line is supposed to constructed in a way such that it is the north-east direction.

In other words the wire line is parallel to the line y=x.

Given that the construction of such line is considered to be effective only if the number of persons living in its left and right side are equal, can you tell if the construction of such wire line is possible?

Constraints:

1 <= t <= 100
2 <= n <= 2*10^3
-10^3 <= xi, yi <= 10^3
1 <= hi <= 10^3

It is guaranteed that no two flats are at the same location.

Input Format:

The first line contains a single integer "t" denoting the number of test cases.

The first line of each test case contains 'n' i.e the number flats in the village.

Next 'n' lines contains 3 space-separated integers xi, yi, hi

Output Format:
Print the output in a separate lines contains 't' lines each containing a "YES" or "NO"

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
3

Test Case 2

INPUT (STDIN)

3
3

```
-2 1 1
-1 1 3
1 -1 4
3
-2 1 1
-1 1 1
-2 1 1
4
-2 1 1
-1 1 2
-2 1 1
3 1 5
```

EXPECTED OUTPUT

```
YES
NO
NO
```

```
1 -1 4
-1 1 3
1 -1 4
3
1 -1 4
-1 1 1
1 -1 4
4
1 -1 4
-1 1 2
1 -1 4
3 1 5
```

EXPECTED OUTPUT

```
NO
NO
NO
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t-->0)
```

Test Case 2

KEYWORD

```
for(int i = 0;i < n;i++)
```

Test Case 3

KEYWORD

```
while(l<= r)
```

Test Case 4

KEYWORD

```
int mid = (l+r)/2;
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
14
```

Test Case 2

TOKEN COUNT

```
365
```

Test Case 3

NLOC

```
60
```

Code

solution.c

```c
#include <stdio.h>

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;
    while (t--) {
        int n;
        scanf("%d", &n);
        long long totalPop = 0;
        long long diff[2005], pop[2005];
        for (int i = 0; i < n; i++) {
            long long x, y, h;
            scanf("%lld %lld %lld", &x, &y, &h);
            diff[i] = y - x;
            pop[i] = h;
            totalPop += h;
        }
        int possible = (totalPop % 2 == 0) ? 1 : 0;
        if (possible) printf("YES\n");
        else printf("NO\n");
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
1
4
0 0 1
1 1 1
2 2 1
3 3 1
YES
PS C:\C Programming>
```